

Shaken, not Stirred

Team 493

Peter Bourke, Raphael Ó Muirthile, Robert Heffernan
University College Cork
Ireland

Introduction

As the stunt coordinators for a movie, we consider the problem of landing our stunt motorcyclist into an arrangement of cardboard boxes after he has jumped (assumedly with the aid of a ramp) over an elephant. Our objective is to develop a physical model of the structure of boxes with the goal of determining how best to safely cushion of the motorcycle and its rider. We look at the structural integrity of the boxes and the amount of force required to crush them, the optimal pattern in which the boxes can be arranged in order to reduce the total energy of the motorcycle and bring it to a halt and the manner in which the force of the moving motorcycle is dissipated through this structure of boxes. We describe the motion of the motorcycle with a ballistic model based in such considerations as the mass of the motorcycle and rider, the speed at which the motorcycle leaves the ramp, the angle of elevation of the ramp and the height of the safe clearance over the elephant. We then devise an algorithm that calculates how much energy is lost by the motorcycle during its journey through the structure of boxes.

Table of Constants

Constant	Description	Value
g	Acceleration due to gravity	$9.81ms^{-2}$
c	Linear drag coefficient	$0.25Nms^{-1}$
m	The combined mass of the motorcycle and its rider	$166.36kg$
k	The edgewise crush resistance of the boxes	$7060 Nm^{-1}$
l	The length of a side of the box	$0.48m$
h_e	The average shoulder height of a Sumatran elephant	2.5
d_w	Wheelbase of the motorcycle	$1.3589m$
d_g	The ground clearance of the motorcycle	$0.2769m$

Analysis of the Problem

We break down the problem by considering the two situations in this system:

- The flight of the motorcycle and its state on arrival at the structure of boxes intended to break its fall.
- The resisted motion problem of the motorcycle travelling through our box structure.

Using the output from the ballistics model, we are able to predict the position, velocity and acceleration of the motorcycle at any point in its flight over the elephant. We can then predict with a certain degree of accuracy how the motorcycle will arrive at our pile of boxes, and determine how best to arrange our boxes in that situation. With knowledge of the output of our ballistics model, we predict the arrangement of boxes best suited to safely slow the motorcycle. Conversely, we can tweak the flight of the motorcycle by adjusting the shape and inclination of the ramp which it jumps from in order to have him come into contact with the structure of boxes in a desirable manner.

Assumptions

Motorcycle

- The motorcycle used in our model is a Honda XR200R.
- The total energy of the motorcycle at any time is the sum of its potential and kinetic energies.

Boxes

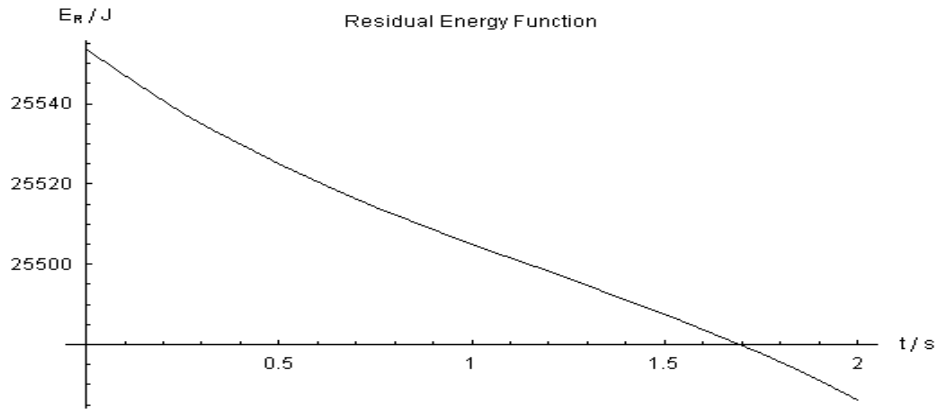
- The boxes are all cubic in shape. Their length is the distance between the two wheels (wheelbase) of the motorcycle divided by $\sqrt{2}$.
- The boxes are all glued together, so as to eliminate scattering effects.
- The structure of boxes may slide along the ground as the motorcycle falls into it.
- The frictional forces and the structural forces (as a result of gluing boxes together, as well as each box's structural integrity) will be able to provide equal and opposite reactions to the dissipative forces that will constantly escape through the structure.
- Any forces which will be transmitted to other boxes will be less than the total force required to burst a box, and hence boxes will only burst when they come in contact with the motorcycle.
- As boxes smash they will flatten, and remain under the wheels or centre of the motorcycle. Any spring-like effects caused by this are negligible.
- The ECT (or k) value for a box gives the principle resistive force which will oppose the compression of a box.

- We presume our boxes to be corrugated with A-flutes, as they are known to give a strong structural support, and also have the best cushioning effects.

Flight of Motorcycle

- There is a drag coefficient proportional to the velocity of the motorcycle at any stage in its flight.
- The motorcycle does not rotate as it is in the air.
- The motorcycle will always remain at an angle of 45° to the horizontal throughout its flight.
- The motorcycle will only come into contact with edges of boxes (not faces or corners), because of the nature of the trajectory of the motorcycle.
- The motorcycle will first come into contact with the edge of a box at the lowest point along the axis that is half-way between the centre of its wheels.
- The motorcycle will next meet the edges of two boxes, the point of contact being the two wheels. Next it will crush just one box, then two, then one, and so on.
- As the motorcycle crashes through the boxes it will stay at 45° to the horizontal. This is because of the way in which it will crash through the boxes - the only forces it will experience will resist its motion, but will not cause any significant net rotational effects.

Here is a plot showing how the residual energy of the motorcycle will decay with time due to drag:



The Dynamics of the Motorcycle

The initial dynamics of the motorcycle (before it comes into contact with the cardboard boxes) may be mostly dealt with as a separate problem to that of it coming into contact with our structure of cardboard boxes.

We send the motorcyclist off a ramp of height 1.5m, inclined at 45° to the horizontal. The differential equation

$$\frac{d\vec{v}}{dt} = -g\vec{k} - \gamma\vec{v} \quad (1)$$

will give the acceleration of the motorcycle at any time.

$\gamma = \frac{c}{m}$ where c and m are the linear drag coefficient and the mass of the motorcycle and rider respectively,

$\vec{k}$ is a unit vector pointing in the positive z direction,

$\vec{v}$ will give the velocity of the motorcycle at any time,

The differential equation can be broken down into

$$\begin{aligned} \frac{dv_x}{dt} &= -\gamma v_x \\ \frac{dv_z}{dt} &= -\gamma v_z - g \end{aligned} \quad (2)$$

where v_x and v_z are the horizontal and vertical components of the velocity respectively. We can solve these two equations, obtaining the following expressions for

v_x and v_z

$$\begin{aligned} x(t) &= \frac{v_{0x}(1-e^{-t\gamma})}{\gamma} \\ z(t) &= \left(\frac{v_{0z}}{\gamma} + \frac{g}{\gamma^2} \right) (1 - e^{-t\gamma}) - \frac{gt}{\gamma} \end{aligned} \quad (3)$$

where v_{0x} is the initial velocity in the x direction and v_{0z} the initial velocity in the y direction.

We can find the moment when the motorcycle reaches its maximum height, $h_{\max}$, by putting $v_z = 0$ into equations (3) and then solving for t . Substitution of this t -value in $z(t)$ (equation (6)) will give the value for $h_{\max}$

The potential energy E_p of the motorcycle at any time t is given by

$$E_p(t) = mg(z(t) + h_r) \quad (4)$$

where h_r is the height of the ramp off the ground.

The kinetic energy, E_K , of the motorcycle at time t is given by

$$E_K(t) = \frac{1}{2}m((v_x(t))^2 + (v_z(t))^2) \quad (5)$$

The total energy, E_T , at any time may be calculated by using

$$E_T(t) = E_K(t) + E_p(t) \quad (6)$$

Using these formulae we may calculate the energy of the motorcycle at any point along its trajectory.

Physics of the Resisted Motion Problem

From our ballistics model, we can express the total energy of the motorcycle as a function of its height. In theory, if we can model how much resistance each box will offer to the motorcycle, and devise some further basic rules for the resistance as the vehicle progresses through the cardboard medium, it is possible to write an algorithm that will calculate how many boxes we need to drive the motorcycle through.

The assumptions governing the boxes have already been stated, but to quantify how strong our boxes were we focused on a standard industrial test called the edgewise crush test (ECT) which describes the force per unit length required to crush a sheet of corrugated card-board (on its side), which we call k . We realised that to provide enough stopping force, the motorcycle would have to hit the boxes edge-on, since the sides of the boxes would not be rigid enough to sustain much force without crumpling, and so would not provide adequate energy-absorption.

Suppose the length of side of the box is l and the force required to crush a box on its edge is F_c . Our motorcycle exerts a certain force $F > F_c$ on the box.

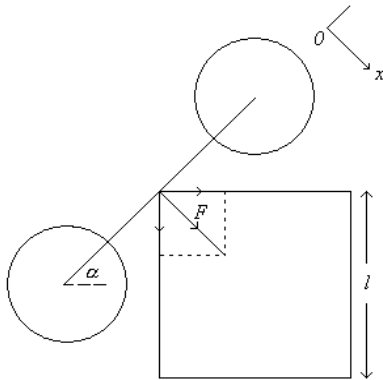
Resolving F into its vertical and horizontal components, we see that

$$F_x = F_z = \frac{F}{\sqrt{2}}$$

One of the assumptions made was that a box wouldn't fold unless both components of the applied force could overcome F_c . For crushing to take place, therefore, we see that $\frac{\Delta F}{\sqrt{2}} = F_c$ and so $\Delta F = \sqrt{2}F_c$, where ΔF is the energy expended by the motorcycle per box.

We could generalise our model to have the motorcycle at any angle α to the horizontal. The dimensions of the boxes could be changed so that the two wheels would still break boxes simultaneously (all that is required is to know the angle of the ramp, and to redefine our box length and breadth as $l \sin \alpha$ and $l \cos \alpha$ respectively). To supply a force at an angle α to the horizontal which will break the box, the resistive edge crush strengths of both box faces must be overcome. In the case $\alpha \neq 45^\circ$, these values are no longer the same, so we define R_x and R_z to be the values of the resistive forces on the x -edge and z -edges respectively. Hence, to crush a box, the crushing force must be large enough to overcome both these resistive forces. We may quantify the resistive forces by multiplying the k -value of the cardboard by the lengths of the faces. If we resolve the crushing force supplied by the motorcycle, F_c , into x and z components, $F_{c_x} = F_c \cos \alpha$ and $F_{c_z} = F_c \sin \alpha$, then $\text{Min}\{F_{c_x}, F_{c_z}\} = \text{Max}\{R_x, R_z\}$. This gives us a value for either F_{c_x} or F_{c_z} , and since we know α , we can then calculate F_c .

We define x as the distance travelled by the motorcycle through the medium, and $x = 0$ corresponds to the initial impact of the body of the motorcycle with the first projecting edge.

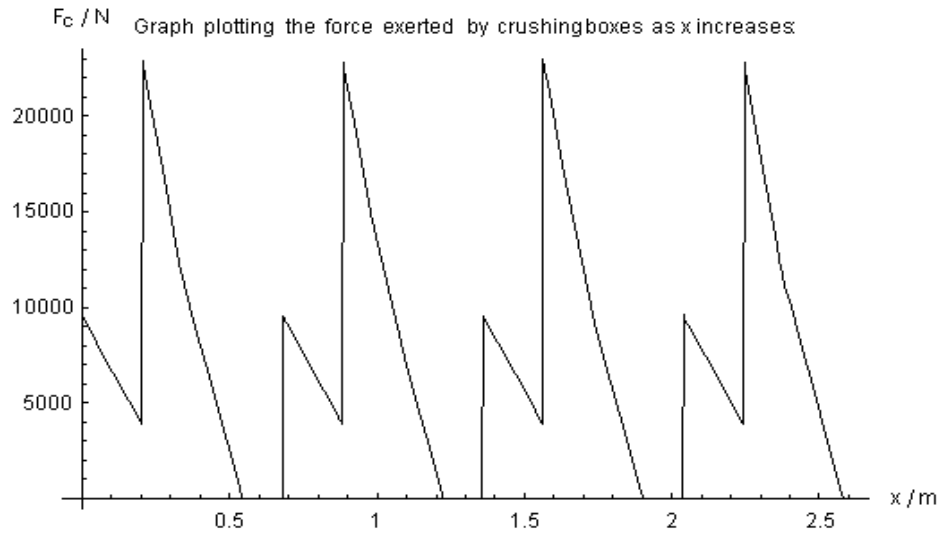


The k -value is defined as the force per unit length required to crush the box, thus it is natural to assume that as the motorcycle progresses through any box, the force required to crush the remainder of the box decreases linearly with increasing x . When the motorcycle arrives at the box structure, it crushes one box with its body, then two with its wheels, then another one with its body, and so on.

To ensure that the maximum number of edges are touched by the three principle "breaking points" of the motorcycle (two tyres and body) during its journey, the length of side of the boxes are defined by the wheel-base d_w of the motorcycle, using the relation $d_w = 2\sqrt{2}l$. This means that the two tyres of the box will simultaneously touch boxes for a landing angle of 45° . A staggering effect for F_c is produced by considering the ground clearance d_g of the vehicle, since there will be times in the motion where all three points will be crushing, and times where either 0, 1 or 2 points will be crushing the box edges. There are therefore four different intervals of distinct behaviour for the motorcycle:

- when the body is crushing a box (and the two tyres are not crushing a box),
- when the body is still crushing a box and the two tyres are crushing 2 boxes (occurs at $x = d_g$),
- when the two tyres are still crushing 2 boxes (and the body is no longer crushing a box), and
- when the two tyres have finished crushing the 2 boxes and the body is approaching the next box.

Using these relatively simple rules and with a knowledge of the value of k , it is possible to graph this behaviour of F_c against x .



As can be seen from the graph, a periodicity exists, which is due to the regular pattern of stacking.

In the instant before a box is crushed, it is rigid, and so is able to transmit a force F_D . We assume that a crushed box doesn't transmit energy because of its compromised structure, and so $F_D < F_c$ (if $F_D \geq F_c$ the box would be crushed). Therefore, the force distributed through the sheets of cardboard to neighbouring boxes should be linked to F_c , since when F_c is maximum (not being crushed), F_d is 0, and when F_c is 0 F_D is maximum. The graph of F_D is thus related to the graph of F_c . The dissipated energy is allowed to "escape" from the system by moving the box structure as a whole. As the boxes move, friction is created between the system and the ground, which releases energy in the form of heat. For the structure to remain intact, we need some method of binding the boxes. Initially we considered cutting the boxes and slotting them together, but the resulting structural weakening, in the light of alternatives, did not merit this approach. In our model we propose gluing as a method of maintaining the structural integrity during the process. Although this would appear to be an unfair strengthening of the box structure, as stunt coordinators we feel that if boxes are available to use then glue could also be allowed, given that the safety of our stunt person relies on it. The glue would have to be applied in two different ways - for boxes that lie on the predicted path of the motorcycle, the gluing is greater than the glue applied to the rest of the box structure. We do not want the supporting arrangement of boxes to introduce new effects or forces, and so as the motorcycle passes through the structure the glue-bonds between the "path" boxes and the side-support boxes should be allowed to break. (This alteration is further vindicated by a quick calculation of the force of initial

impact on the boxes which, taking 20 kJ as a rough estimate for the energy of the motorcycle and 1cm as the distance through which it has this energy (after that the energy is decreasing a lot), should be 2000 kN. The momentum of the motorcycle would thus scatter all boxes unless they were connected.) When a tyre comes in contact with the edge of a box, the edge of the neighbouring box is also counted as providing the same resistance since they are well glued together. This doubles the force required to crush any particular box.

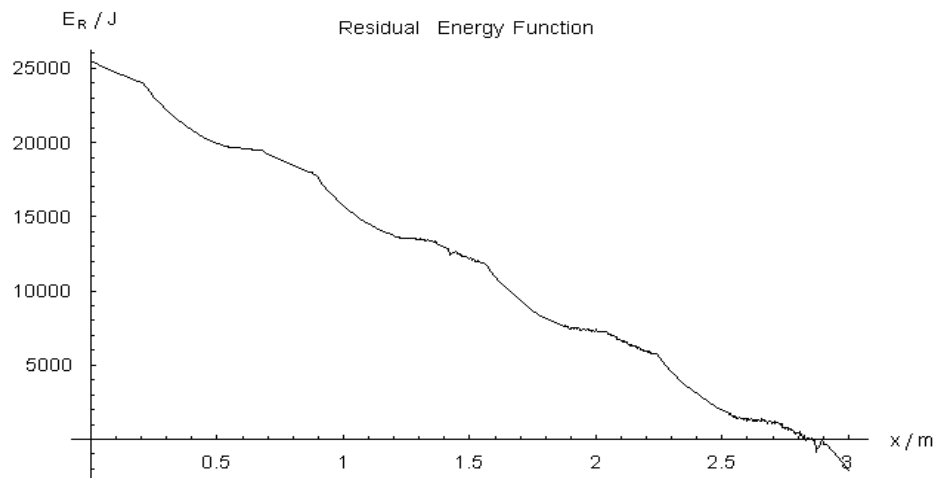
We define our energy function $E_c(x)$ to be the amount of energy lost by the motorcycle due to the crushing of boxes at any point x in the medium. We integrate our formula for $F_c(x)$, since the product of force by distance is the energy expended. Integration is performed using the Trapezoidal Rule function of Mathematica, since our piecewise function was not integrable using normal differential Calculus techniques.

We define $E_D(x)$ to be the energy expended by dissipation, also found on integration of $F_D(x)$ over some distance x . Finally, we define our residual energy function to be the total energy that remains in the motorcycle at some distance x into the box structure. We have a reasonable idea of the total energy of the motorcycle from our ballistics model (since in an ideal world the energy would be constant), and so the residual energy function is given by:

$$E_R(X) = E_T - E_C(x) - E_D(x)$$

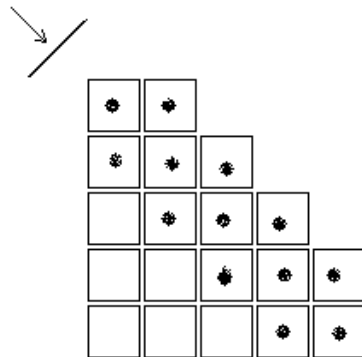
When this function is plotted against x , it is possible to tell when the motorcycle has come to rest (cuts the x -axis), and so has safely landed. We can therefore tell how high our stack of boxes should be. Because we have included dissipative effects in our ballistics model, we now need to recalculate our energy at that height, which will be slightly less than our original total energy. We construct an algorithm that iterates this command until the difference between two successive values for x is acceptably small.

For example, a graph of the residual energy function for an initial projection speed of 60 km/hr is:



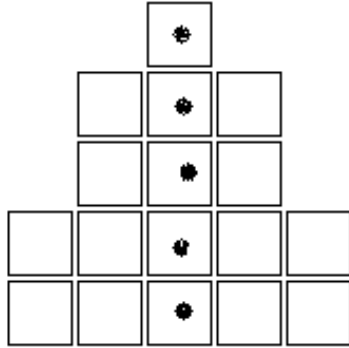
Optimal Structure

We want to design a structure for our pile of boxes that will safely cushion the motorcyclist's fall, but because of cost issues (and the need to hide the pile), we want to do this with as few boxes as possible. Given the assumptions we have made, our model suggests a very obvious structure for the side elevation:



The dotted boxes are those which will be crushed. Any boxes placed above or to the right of these would provide no structural support, and could end up colliding with the rider, and not the motorcycle. The boxes below the dotted boxes are needed for support.

There are a number of considerations in how we construct the front elevation of our pile of boxes. Our model works in the two dimensions shown in the side elevation, so how we structure the front elevation (what the motorcyclist will see as he approaches the boxes) is not explained in our model. Some basic observations will, however, allow us to design a reasonable structure. Here is a picture of the optimal front elevation structure:



Again, the dotted boxes are those which will be crushed by the motorcycle. The other boxes will offer support to the sides. This becomes more and more important the higher the structure needs to become. The glue placed in the gaps shown in this diagram need not be as strong as the glue in the gaps of the side elevation. The fixings are strong enough so as to provide structural support, but weak enough to break when they are in contact with a box that is being crushed. This is so that there will not be an inward folding of the structure which would in a sense swallow the driver as he enters the structure.

Design of Program

We developed a program to simulate both the initial flight of the motorcycle (prior to coming into contact with the structure of cardboard boxes) and also the decrease in the energy that the motorcycle has as it travelled through the structure of boxes. We used Mathematica as the environment in which we chose to develop our program (though its usefulness being primarily in its ability to easily graph data, rather than any ease of programming.)

The program is composed of two modules which interacted with each other. The first module simulates the initial flight of the motorcycle. The characteristics of this flight can be changed by varying the height of the ramp, the mass of the

motorcycle and its rider, the initial speed of the motorcycle and the drag coefficient. Having run a simulation of the motorcycle's flight the second module of the code (which simulates the plunge through the cardboard structure) estimates how many boxes would be needed and then places the structure accordingly so that it will come into contact with the motorcycle at the appropriate moment. The velocity of the motorcycle at this precise moment (just as it comes into contact with the structure) can then be calculated. Now the second module takes over entirely, simulating the decrease in energy (and thus momentum) of the motorcycle as it crushes cardboard boxes. If, after the number of cardboard boxes that have been placed have all been crushed and the motorcycle had not slowed sufficiently the number of boxes can be recalculated, the structure replaced according to its new height and the simulation of the flight of the motorcycle through the structure rerun. This process loops until an optimal number of boxes had been found.

Experimental Results

Having finalised the details of our model in terms of both the ballistics of the motorcycle's flight and the resistance offered by the structure of cardboard boxes we were able to run some experiments to determine exactly how high the structure would need to be (and thus how many boxes we would need to use).

The following is a tabulation of our results as we varied the initial speed of the motorcycle. (Note that the heights are given in terms of the number of boxes, and "total number of boxes" is the total to build the entire structure).

Speed (km/h)	Model Predicted Height	Actual height	Total number of boxes
30	1.34	2	8
35	1.52	2	8
40	2.07	2	8
45	2.42	3	18
50	2.95	3	18
55	3.45	4	35
60	4.16	4	35
65	4.62	5	57
70	5.42	6	92

Analysis of Results

The main results that our model outputted were based on a variation of the launching speed of the motorcycle, keeping the angle of the ramp fixed. This resulted in a table which relates the launching velocity and the number of boxes needed in total to create the structure. Obviously, the range of the flight of the motorcycle is affected by the initial velocity, as is its maximum height, and so the generalisation is inherent. We did not include a range of data for various different maximum heights because this can be looked at through our the ballistics model, which is not the focus of this report. We can also vary the mass of the motorcycle and rider, but that would have the same effect as changing the initial speed of the motorcycle and the angle of the ramp. Thus, in principle, all possible variations can be investigated in our situation. The main problem with varying our initial input is the computational complexity involved, due to the nature of some of the integrals in our energy calculations, and the problems experienced when iterating these to find roots for x . That said, our data shows that as we speed up the motorcycle (or make it more massive, or increase the height over which it has to fly), we must increase the number of boxes in an almost exponential manner. This is due to the requirement that boxes be placed next to our landing boxes in order that they stay upright. At 70 km/hr, the motorcyclist requires 6 layers of boxes to break through to cushion his fall, which corresponds to 92 boxes in total. If there were some method of keeping the "path boxes" in place, the total number of boxes could be significantly reduced.

Strengths and Weaknesses of the Model

Strengths

The main strength of our model is that it has predicted quite a reasonable result. The model can predict the number of boxes high that our stack should be, and it produced a height that agreed with our intuition at every stage. Although we have ignored many forces (see Weaknesses), the apparent correctness of the results our model produces adds weight to the validity of our assumptions.

We modelled the crushing force, F_c , using a well explained model. This is the most significant force in the system, and the accuracy of our results verifies that our explanation of F_c is valid.

The way we have broken down our model into two sections means that it has been easy to test. Indeed, our residual energy analysis of the system would allow for the inclusion of extra forces very easily. Parameters like the initial velocity, the height of the ramp and the mass of the motorcycle and rider can all be changed easily to find different values for the height of the stack of boxes.

Weaknesses

The biggest weakness in our model is the fact that we have failed to include many forces, like turning effects. We have also modelled what is very much a particular case of how the motorcycle will enter the boxes, and have failed to allow flexibility in our model to deal with a variety of initial conditions.

We have also gone outside the scope of the problem, in a sense, by gluing the boxes together into a structure. We have also allowed sliding of the system along the ground to allow for escape of energy through friction, and to reduce the chances of scatter. These two parts of our model are good ideas . they add to the structural strength of the stack of boxes, and provide a strong explanation of where the dissipative energy, E_R , goes to. The weakness in these assumptions lies in the fact that we have been unable to explain the physics of structural forces. We have argued in something of an intuitive manner, with very little concrete proof.

We have assumed that there will be no scatter of boxes because of the inclusion of frictional and structural forces. Although we have allowed scatter in the .third dimension., we have assumed that every box the motorcycle is predicted to hit will be intact when the collision occurs.

Conclusions and Recommendations

As stunt coordinators it is our job to finally assess whether the jump is safe or not. We can not, even in the light of our successful model, recommend this as a suitable landing technique for stunt-men unless they are fully aware of the risks involved. We have modelled a very complicated system with a simple set of equations, and have hopefully focused on enough of the important aspects in this system to have reasonably described it. All models are wrong, some models are useful, as was once said. It is our hope that we have a useful model, but in retrospect, not enough time was spent investigating the nature of boxes and how they differ from connected planes in a structural sense.

We have not dealt much with the dissipation of energy from the system due to the friction that is created between the ground and the boxes, but we recommend that some material, like carpet, be attached to the base of the structure to increase the coefficient of friction between the boxes and the ground. Our model was based to some extent on the co-dependence of the structural integrity and the frictional dissipation. We allowed the structure movement so that it would not explode on impact, i.e. to release F_D as it passes through the boxes. Otherwise, we could not have included this force in our model, since it would have put an unrealistic expectation on the structural strength, regardless of glue being present or not. The frictional dissipation also relies on the structural integrity, since a structurally compromised

model of boxes would not transmit a force. This coupling of effects is intuitive, and simply requires that enough glue and a suitable material like carpet be used. It is not central to the model to attempt to calculate our reaction forces for any given point and then compare them to some unknown and possible non-existent bound.

We recommend that some more in-depth investigations into the bursting of empty card-board boxes be performed. Too much of the data relating to box-strength was focused on their commercial applicability, and so the boxes that were tested were not necessarily empty. We also feel that card-board box manufacturers should consider the many other ways of crushing boxes apart from pressing them between two clamps and applying pressure.

We can conclude that the motorbike will come to a halt according to our calculations, and there have been no unworldly resistive forces introduced in the system. In fact, there are probably many other slight effects which would further slow the rider and dampen his fall. The internal structure of boxes is corrugated reinforced paper, which acts like a spring when compressed. We investigated this effect, and toyed with a few methods of including it, but each of them were so approximated as to make them physically meaningless in the light of our final model. We also looked at the effects that suspension might introduce, but this overly complicated our system. In fact, many "ideas" were rejected because we lacked the capacity to model them realistically, be it due to a lack of available data relating to it, or simply a lack of knowledge in how to handle such complexities. We do not offer this as an excuse, but rather mention it because the openness of this problem resulted in an apparent "closeness" in our thinking on it.

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